CONTINUOUS DELIVERY / TRUNK-BASED DEVELOPMENT

FEEDBACK LOOPS MUST BE SHORT

The longer it takes to discover a change broke something, the more expensive the fix. Automated tests, feature flags, canary deployments, and staged rollouts compress the feedback window from days to minutes. Trunk-based development — everyone merging to main daily — is the prerequisite.

KEY BENEFITS

- ✓ Bugs found in minutes cost a fraction of bugs found by customers days later
- ✓ Canary deployments limit blast radius — bad releases reach 1% before 100%
- ✓ Feature flags enable instant rollback without a full deployment
- ✓ Frequent merging prevents long-lived branches that diverge and break on merge

REAL-WORLD EXAMPLE

Netflix deploys hundreds of times per day. A problematic release reaches 1% of users before full rollout — a near-instant feedback loop that bounds the blast radius of any mistake.

PRACTICAL EXAMPLE

A new recommendations algorithm ships behind a feature flag to 2% of traffic. Engagement metrics drop 8% in that cohort. The flag is turned off within 20 minutes. Zero rollback ceremony. Zero user impact at scale.

REFLECTION QUESTION

How long from code commit to knowing whether it broke something in production?